

Digital Infrastructure

Global Financing Trends And The Asia-Pacific Outlook

Live Webinar And Q&A – Friday, October 16, 2026

S&P Global
Ratings

Speakers



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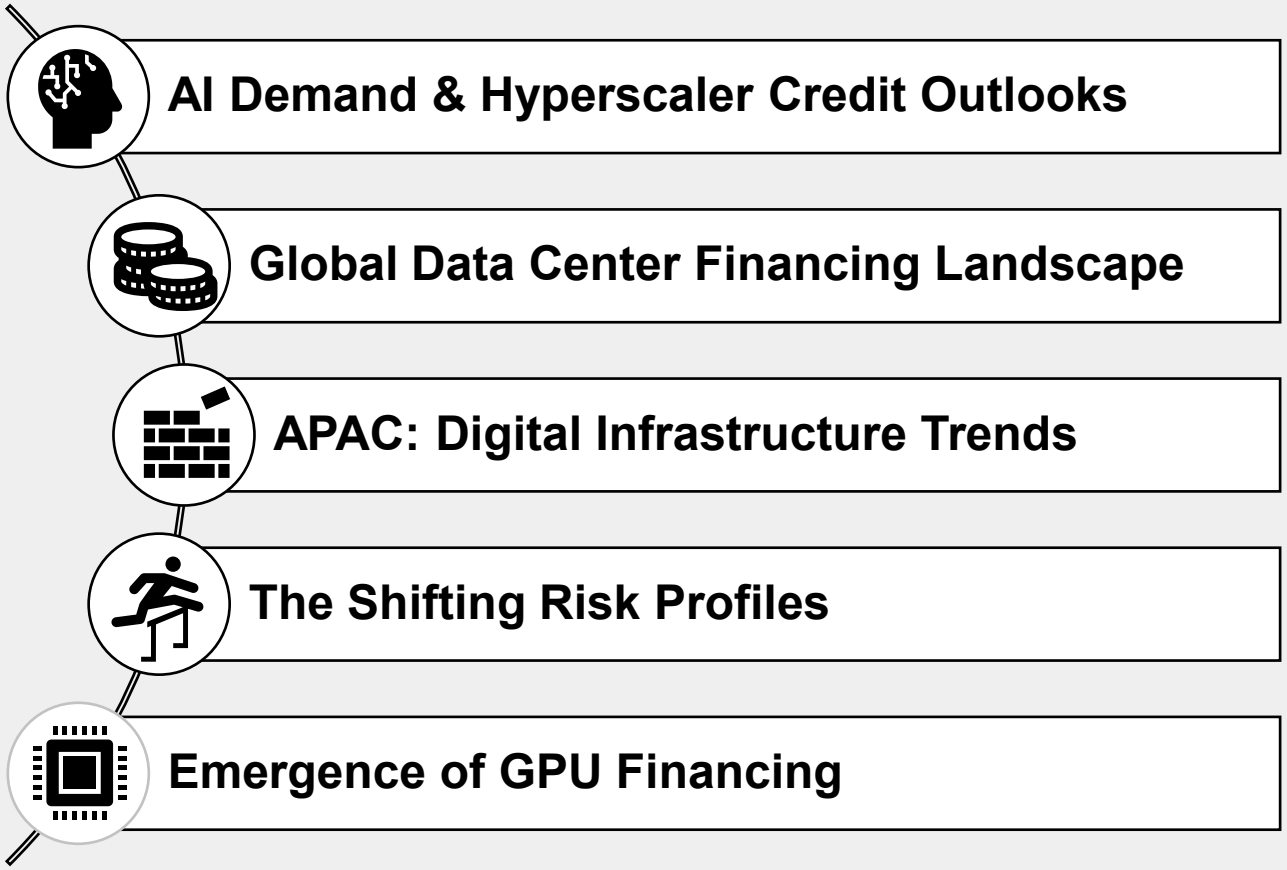


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Agenda

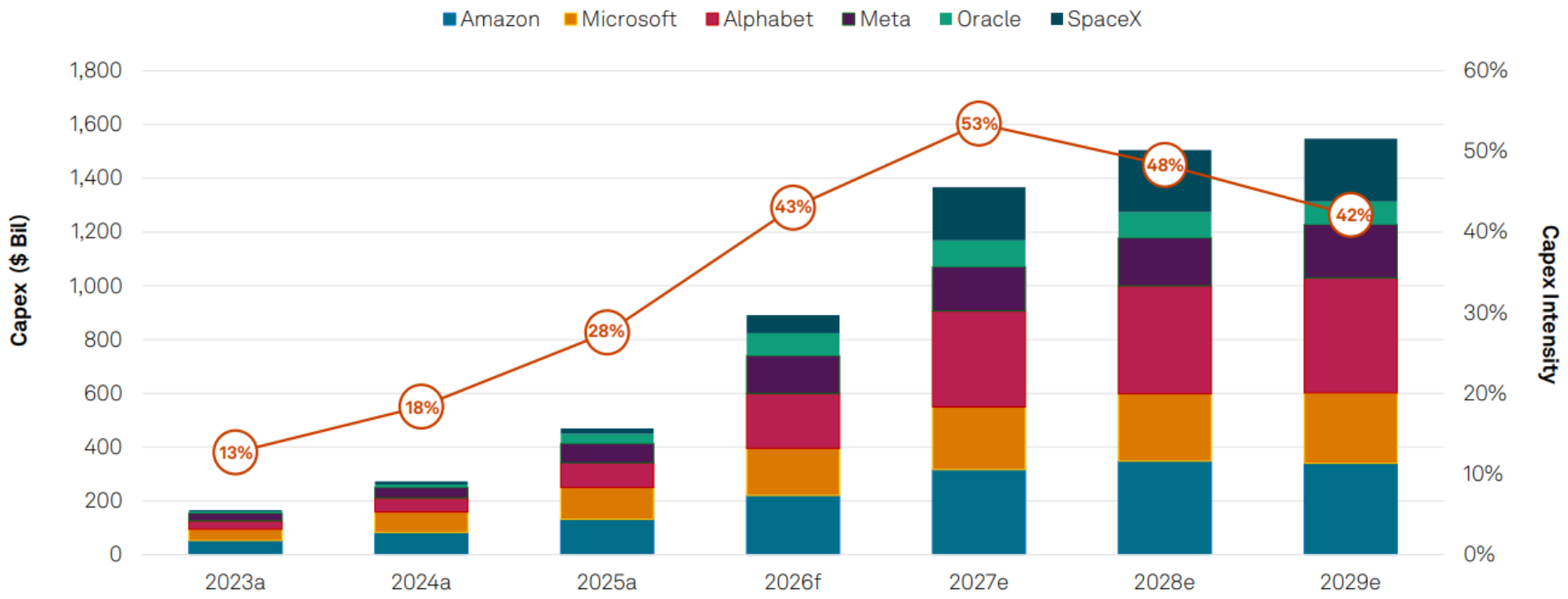


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AI Demand & Hyperscaler Credit Outlooks

AI Demand Signals Remain Strong

- Capex spending will exceed $1.3 trillion in 2027, versus $870 billion in 2026 and $470 billion in 2025.
- Strong demand backed by their own internal workloads (i.e. MSFT CoPilot) and multiyear contracts from AI labs (OpenAI, Anthropic), AI native companies, and enterprises.



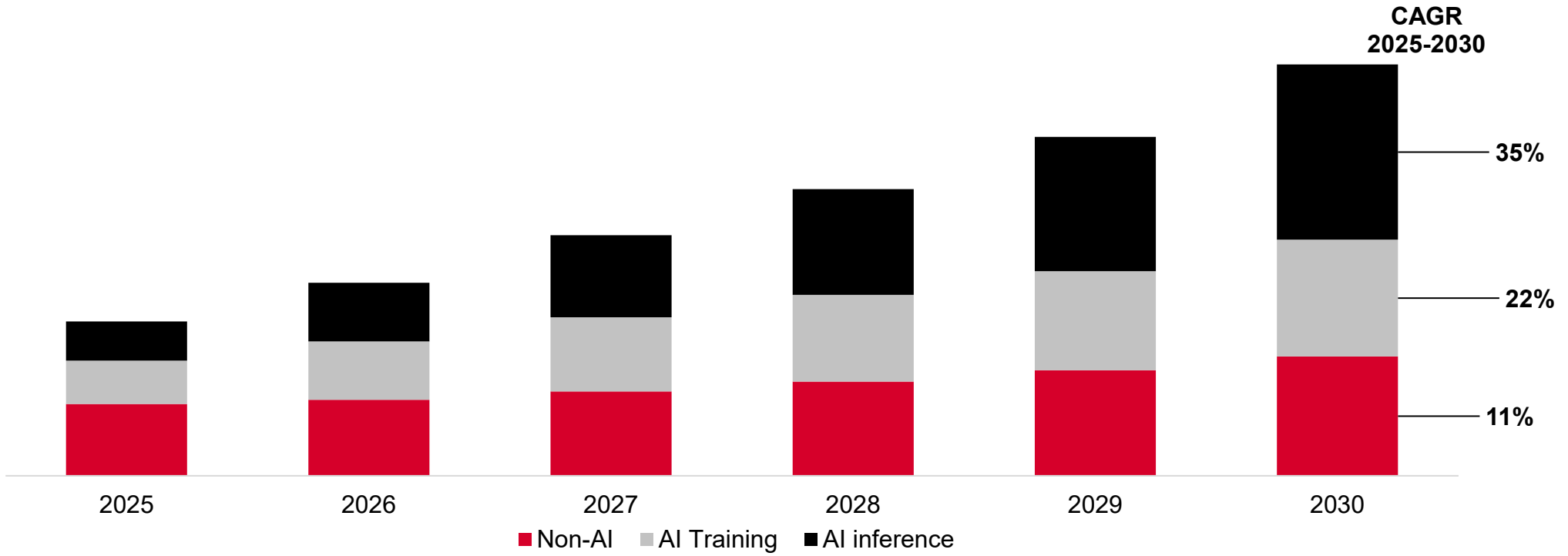
Source: S&P Global Ratings.

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AI Demand Is On Top Of Traditional Cloud and Colocation

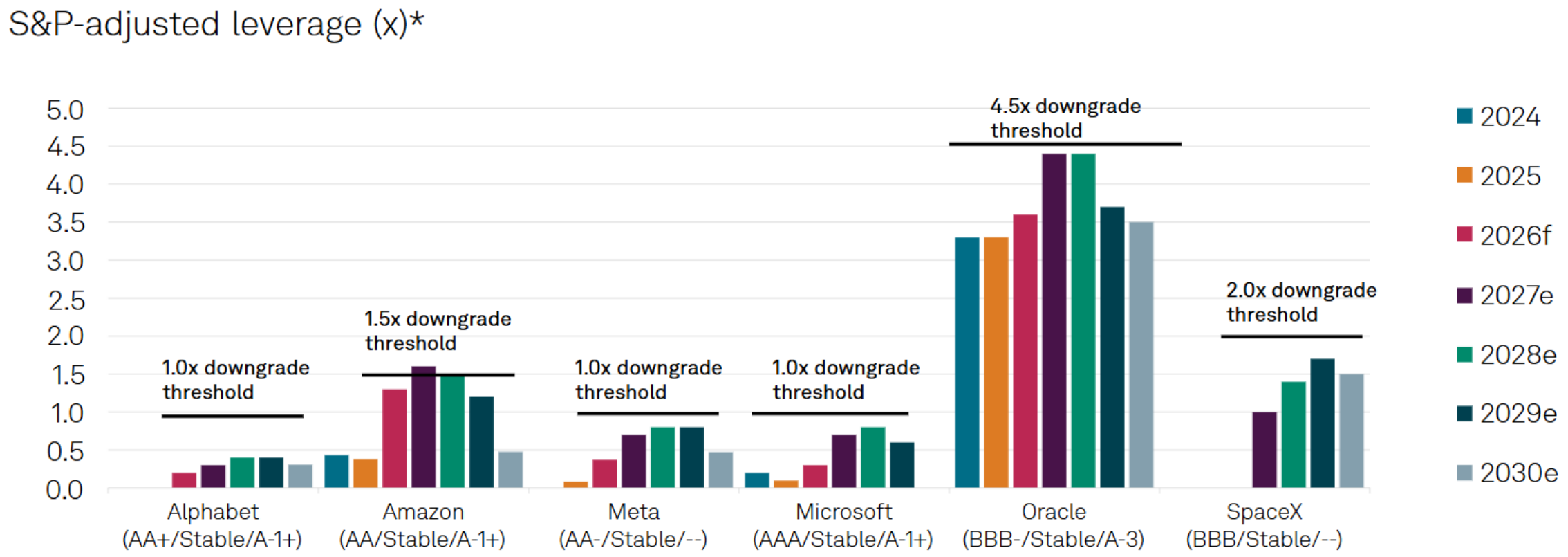
- Hyperscalers' AI training demand will pivot to inference in the coming years.

Global data center demand by workload, 2025-2030



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Leverage To Peak In 2028 As Monetization Grows And Capex Flattens



*Debt to EBITDA. Microsoft's fiscal year is to end-June, Oracle's is to end-May. SpaceX doesn't show pre-IPO leverage (2024 and 2025). Lines show downgrade thresholds. f--Forecast. e--Estimate. Sources: Company filings, S&P Global Ratings.

Hyperscalers' Off-Balance Sheet Debt Is Rising

$ in billions	Alphabet	Amazon	Meta	Microsoft	Oracle	SpaceX
Rating	AA+/Stable/A-1+	AA/Stable/A-1+	AA-/Stable	AAA/Stable/A-1+	BBB-/Stable/A-3	BBB/Stable
LTM period ending	June 30, 2026	June 30, 2026	June 30, 2026	June 30, 2026	May 31, 2026	June 30, 2026
Revenue	445.9	775.7	228.2	331.8	67.4	23.1
EBITDA	205.1	180.5	137.8	213.6	37.5	9.8
Capex	131.5	173.5	88.7	115.9	55.7	43.1
Free operating cash flow	56.3	(0.3)	45.3	73.1	(22.1)	(34.2)
Cash & marketable securities	162.5	122.9	90.3	76.7	31.9	100.0
S&P Global adj. debt	4.6	190.2	28.7	52.2	136.4	0.0
S&P Global adj. leverage (x)	0.0	1.1	0.2	0.2	3.6	0.0
Downside threshold (x)	1.0	1.5	1.0	1.0	4.5	2.0
Debt capacity	200.6	80.7	109.2	161.5	32.4	78.5
Future lease obligations	85.2	137.2	279.0	329.1	260.0	1.6

Note: All figures and metrics are S&P Global Ratings adjusted except future lease obligations Debt capacity defined as excess debt needed to reach downside threshold as of today. Does not include impact from future lease obligations. Sources: Company reports, S&P Global Ratings.

Global Data Center Financing Landscape

Different approaches to rate data center transactions

Rating approach largely hinges on structural considerations: asset type is not the primary differentiator

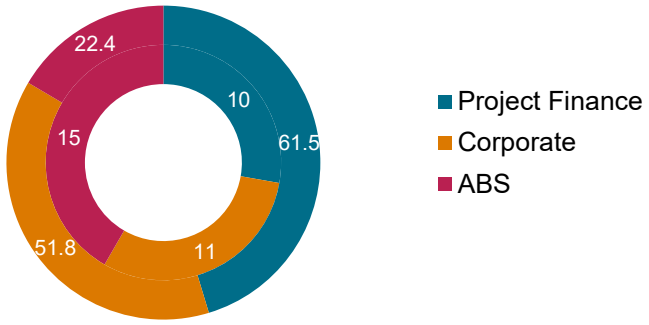
	Corporate infrastructure	Project finance	Asset-backed securities
Issuer	Corporate	Limited-purpose entity	Bankruptcy-remote Special-Purpose Vehicle (SPV)
Construction risk	May be present	May be present	Mitigated or limited credit
Operating risks	Full exposure	Full exposure	Limited (transferable and replaceable)
Renewal risk	Unmitigated (may have full market exposure)	Quantifiable (with impact on market exposure).	Quantifiable (re-leasing risk factored in)
Debt (typically)	Senior secured or unsecured	Senior secured	Multi-tranche, secured
Creditor protections	Typical corporate debt with limited covenants or security	Ring-fenced structure with covenants and security over assets, accounts, and key contracts	Ring-fenced with security over the assets and/or cash flows, and the ability to liquidate and sell
Debt features	Short- or medium-term legal final maturity dates	Medium-term legal final maturity dates	Shorter Anticipated Repayment Date (ARD) with longer legal final maturity dates
Refinancing risk	Allowed--assessment of maturity profile and liquidity	Allowed--reliance on operating cash flows	Expected but not mandatory at Anticipated Repayment Date (ARD) , reliance on both operating cash flows and liquidation value

Data Center Rating Distribution

- We rate 36 transactions (inner circle). Please note for simplicity, we have only considered transactions and not # of tranches which is common approach to measure SF deal volume. Overall, issuance volume surpasses $115 billion.
- Most of the issuances are concentrated in the US market (+75%). There were some issuances also in EMEA, Japan and Canada.

S&P Rated Portfolio

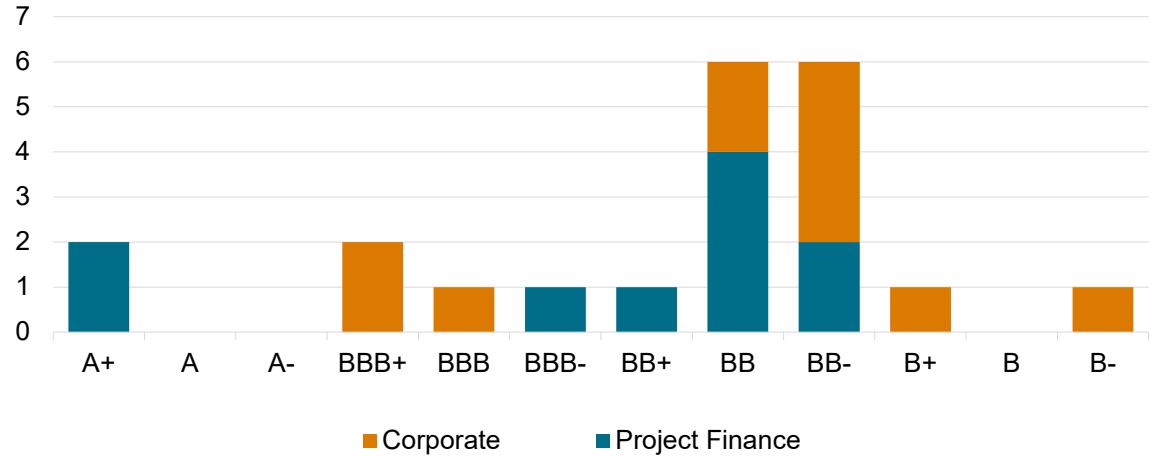
Number of transactions (inner circle)
Rated debt $ Billion (outer circle)



ABS data as of June 2026. Source: S&P Global Ratings.

Rating landscape

Number of transactions (by issue credit rating)



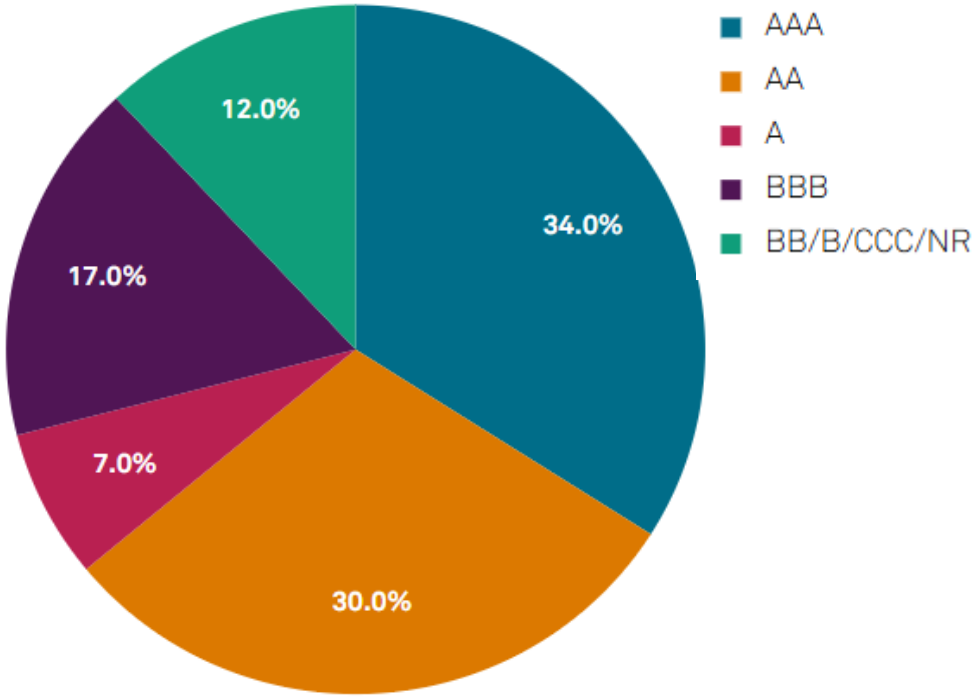
Source: S&P Global Ratings.

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Global Data Center ABS | Ratings Holding Steady Even In Downside

Tenant credit rating exposure by AABR (%)*

Global data center ABS



Data as of first-quarter 2026. *Assumes a spot rate of 1.00 GBP = 1.37 USD and 1.00 euro = 1.17 USD. AABR--Annualized adjusted base rent. ABS--Asset-backed securities. NR--Not rated. Source: S&P Global Ratings.

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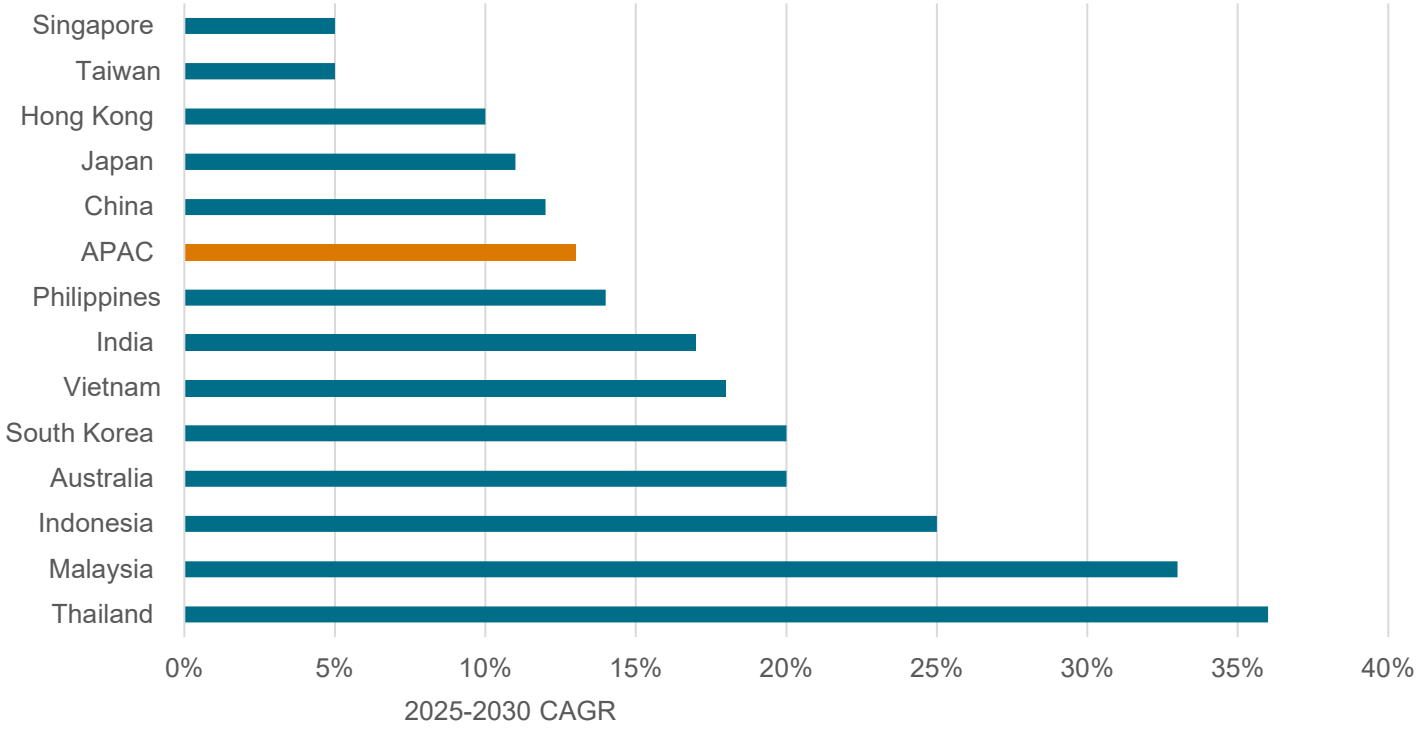
- Data center ABS ratings are resilient to incremental tenant credit migrations and potential increase in vacancy after the initial lease term.
- This resilience is supported by long-term leases and high-quality, mission-critical tenants.
- Despite evolving risks like tenant concentration and lease rollover, data center ABS continues to demonstrate strong structural stability.

Access Article: Global Data Center ABS Ratings Are Holding Steady, May 20, 2026

APAC:
Digital Infrastructure Trends

Asia-Pacific Data Centers | Outsized Growth In Several Markets

Forecast Growth Rates Of Asia-Pacific's Data Center Markets

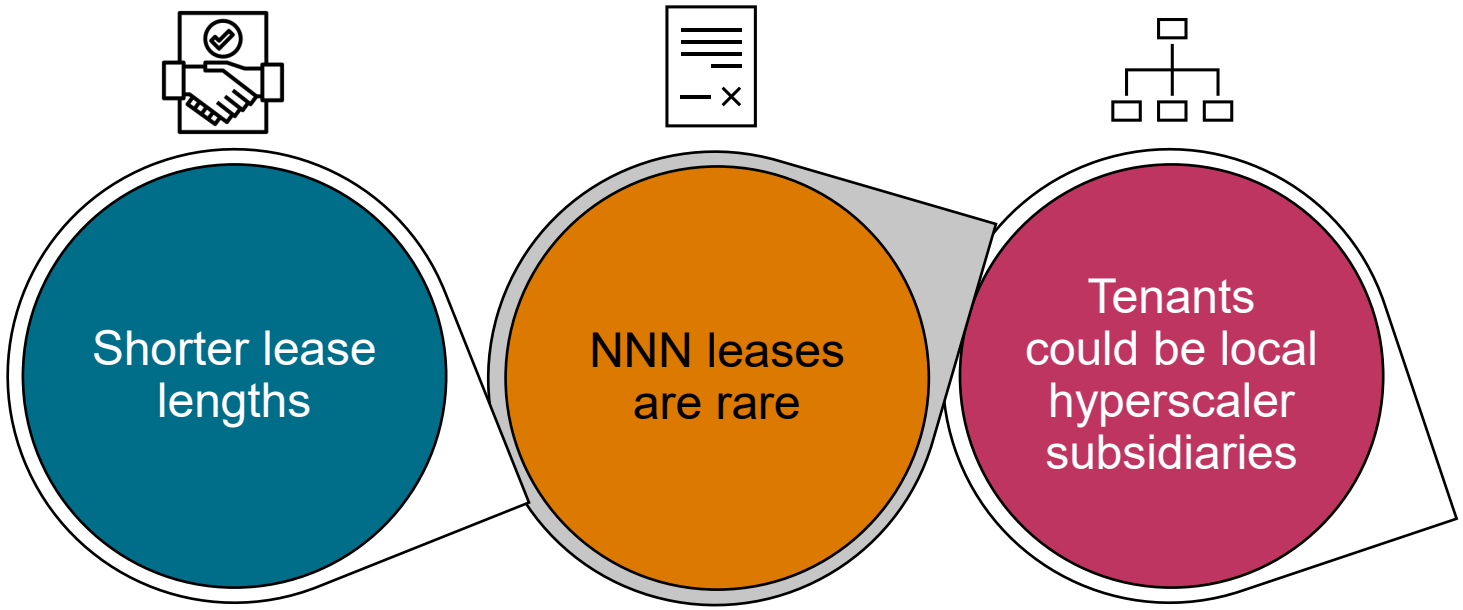


CAGR based on total DC IT power. CAGR—compounded annual growth rate. Source: 451 Research.

- **Indonesia** and **Thailand** will benefit from spillover from Singapore and increasingly, Malaysia. This is on top of strong underlying demand in the local context.
- **Australia**'s strong institutional stability and proximity to Asia positions it well for regional users.
- **Korea**'s strong demand is from global and domestic companies; Korean companies are focusing on physical AI services.
- Power and water resource sufficiency are key watchpoints, with regulatory changes and build freezes of various magnitudes (e.g. Australia, Malaysia, Singapore, Thailand).

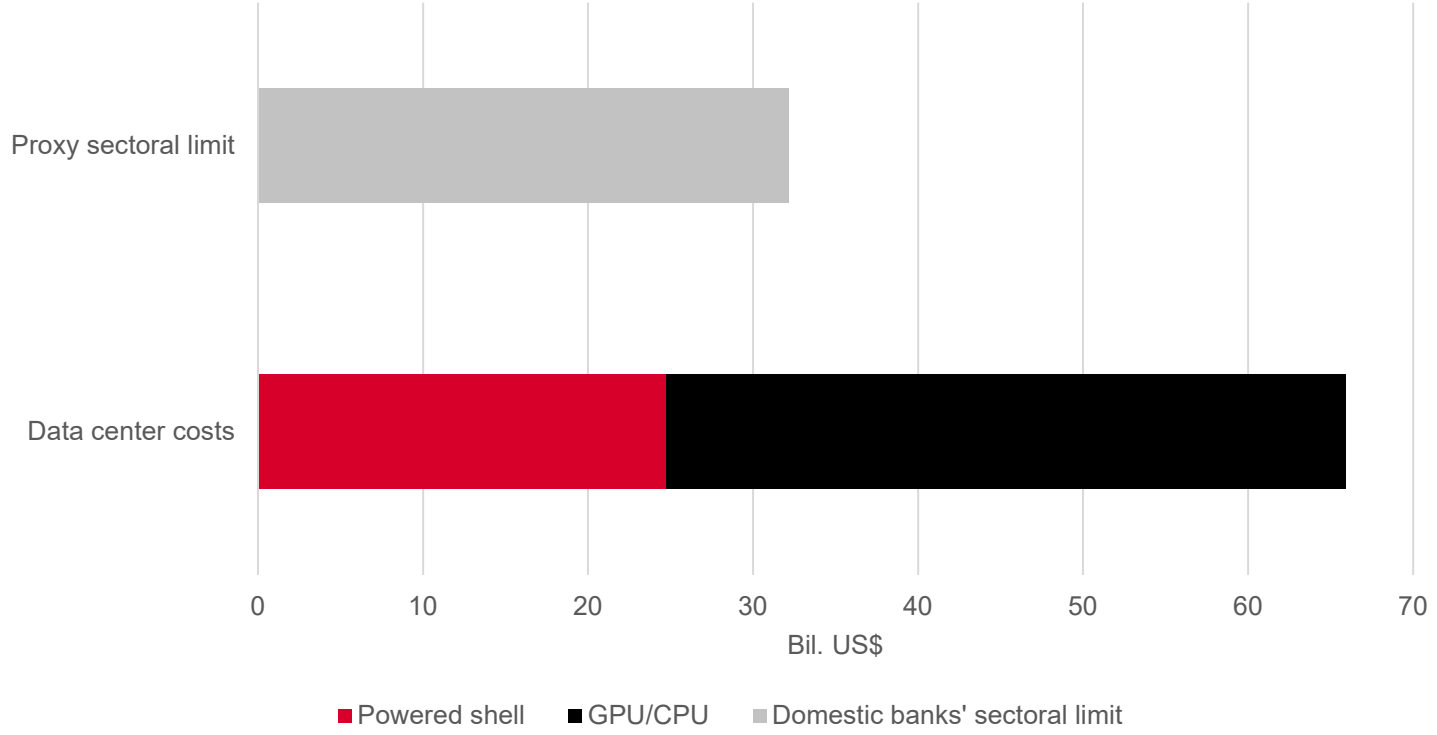
Asia-Pacific Data Centers | Comparison With US Projects

- Data center projects could face reletting risk sooner in Asia-Pacific and be exposed to some level of operating cost variability.
- Tier 1 markets in this region, like in others, will likely offer stronger reletting prospects.
- The counterparty creditworthiness cannot be assumed to be equated to that of the parent.



Asia-Pacific Data Centers | Move From Bank Financing

Malaysia as an example:
Data center costs over the next three years will exceed banks' limits



Note: Proxy sectoral limit is based on public data from Bank Negara Malaysia. It assumes that the single largest exposure is the proxy of the banks' upper bound sectoral limit. As of June 30, 2026, the sector to which Malaysia's banking system has the largest exposure to is real estate activities, totaling MYR130 billion, or close to US$30 billion. This excludes the household sector, as well as the financial and insurance sector. Data center costs assume these are inference data centers. Training data centers will cost more.

- Data centers have mainly been funded by banks and equity
- Anecdotally, we understand banks are hitting their sectoral limits
- We expect **structured finance** to be the first port of call in mature markets with supportive regulations.
- **Project finance and private credit** will bridge the developmental gap.
- Capital recycling via **REITs and funds** could happen later in the data center operator's life.

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The Shifting Risk Profiles

Global Data Center Project Finance | Evolution of Structures

How is **data center** financing being **reshaped**?



2024-2025

DC 1.0
Stable Foundations

- Full construction wraps
- Long-term leases
- No re-letting risk
- Predictable cash flows
- Fully amortizing debt

2025

DC 2.0
Emerging Risks

- Partial pledges
- Shorter lease terms
- Non-IG or NR tenants
- Partial mitigation of construction risk
- Refinancing risk

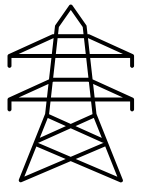
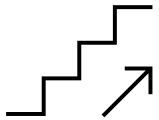
2026

DC 3.0
Elevating the risks

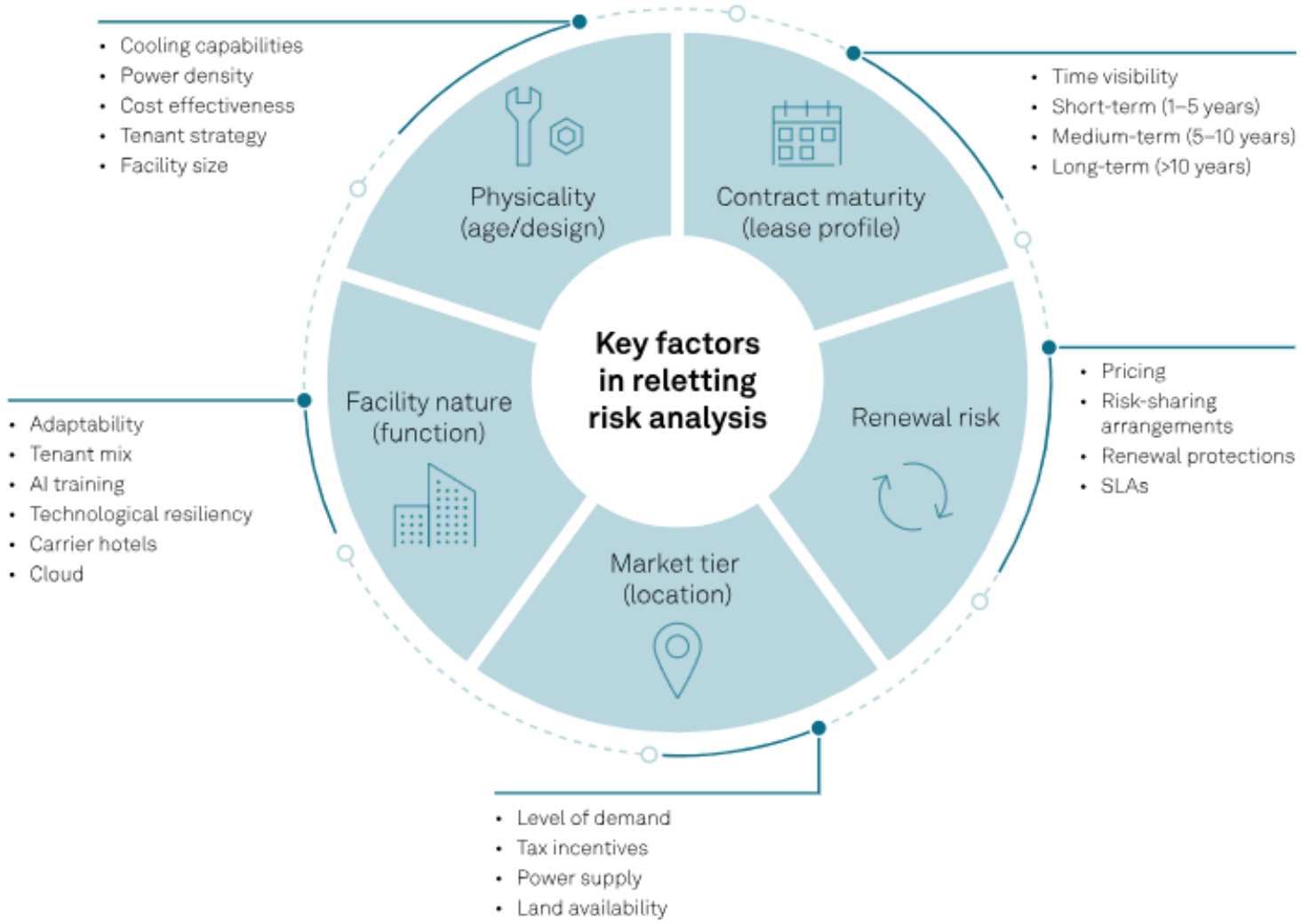
- Longer life assumptions (up to 30 years)
- No rated tenants
- Significant refinancing risk
- No construction wraps
- Power projects included within the scope of the project
- Re-letting risk that generate greater uncertainty in cash flows

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Global Data Center Project Finance | Emerging Credit Trends

Rising Construction Risk	Evolving Lease Structures	Rising Leverage and Refinancing Risk	Insurance Adequacy Concerns	Social & Regulatory Risk
				
Rising scale and scope (i.e. BTM) are increasing execution and completion risks. Deviations from original time schedules and budget.	Evolving lease structures and tenants are driving varying credit implications (i.e. termination conditions and fees, unrated tenants, re-leasing exposures.)	Rising debt levels (LTC) coupled with loosening credit terms. Refinancing rates.	Rising scale, scope and leverage are influencing insurance coverage.	Growing scrutiny regarding the social and sustainability of data center operations (moratoriums/ regulations).

Assessing Re-letting Risk



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Location Is Key For Data Centers

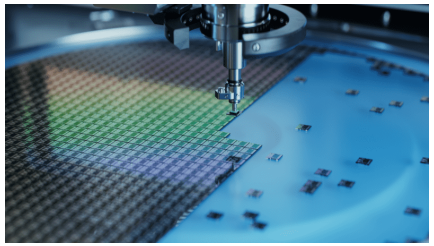
Location is one of the primary determinants of a data center's reletting risk (the real estate factor)

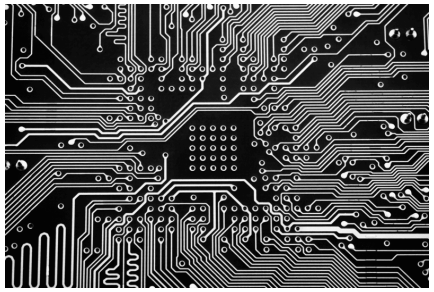
PRIMARY MARKET	SECONDARY MARKET	TERTIARY MARKET
• Business hubs with strong demand drivers (i.e. favorable tax regimes, power constrained) • Low latency • High tenant pools	• Less resilient economic fundamentals • Medium tenant pools	• Remote locations, far from population centers • Higher latency • Fewer prospective tenants

Region	Tier 1 markets	Secondary markets	Tertiary markets
North America	Atlanta, Chicago, D.C., Metro, Dallas/Forth Worth, N. New Jersey, Phoenix, Portland, San Jose, Toronto	Austin, Boston, Columbus, Denver, Las Vegas, New York, Los Angeles, Reno, Richmond, Salt Lake City, Seattle	Louisiana, Utah, North Dakota
EMEA	London, Paris, Frankfurt, Amsterdam, Dublin	Berlin, Dubai, Madrid, Milan, Zurich, Warsaw	Helsinki, Manchester, and Joburg
Asia-Pacific	Tokyo, Hong Kong, Singapore, Sydney, Seoul, Mumbai	Batam/Jakarta, Osaka, Johor/Kuala Lumpur, Melbourne	
Latin America	São Paulo, Mexico City, Buenos Aires, Santiago	Rio de Janeiro, Monterrey, Lima, Bogotá, Medellín, Guadalajara, Porto Alegre, Curitiba, Fortaleza, Valparaiso	

Emergence Of GPU Financing

GPU Financing | Key Credit Considerations





Construction / Installation Risk	• Is installation risk in scope at time debt drawn? • Are components on the site? • Testing and acceptance • Buffer and penalties if delay or ramp-up trouble shooting
Operational Risk	• Generally higher than data centers (specialized maintenance, integrated system, track record hardware) but low overall • Level of hot/ warm / cold spare parts is key • SLAs • Technology obsolescence, specially from year 5 • Power costs variability
Counterparty Risk	• Offtaker • Data center • Equipment and installation / O&M counterparties depending on scope

Comparing Equipment ABS With GPU Financing

Asset characteristics	Typical equipment ABS	Pool of GPUs
Lessee pool diversity	Diversified	Concentrated
Lessee credit quality	Situational	Situational
Equipment useful life	Varies (3-30 years)	Short asset life (4-6 years)
Obsolescence risk	Varies, usually low	High
Equipment maintenance	Lessee responsibility	GPU service provider or data center tenant
Equipment residual value	Limited residual value	Limited residual value due to rapid depreciation
Initial lease term vs. rated debt term	Fully amortizing	Front-loaded amortization to zero or bullet
Location of data center	N/A	Situational
Equipment repossession/redeployment	Low risk	High risk

Appendix

Digital Infrastructure Ratings

Name	Asset Type	Rating Approach	Debt Rating
Beignet Investor LLC	Data Center	Project Finance	A+
Sopaipilla Investor LLC	Data Center	Project Finance	A+
Digital Realty Trust, Inc.	Data Center	Corporate	BBB+
Equinix Inc.	Data Center	Corporate	BBB+
Global Switch Holdings Ltd.	Data Center	Corporate	BBB
Hut 8 DC LLC	Data Center	Project Finance	BBB-
SV RNO Property Owner 1, LLC	Data Center	Project Finance	BB+
Core Scientific, Inc.	Data Center	Corporate	BB
IS AN03 LLC	GPU	Project Finance	BB (Prelim)
PR RNO Property Owner, 1 LLC	Data Center	Project Finance	BB (Prelim)
SE Cosmos, LLC	Data Center	Project Finance	BB
TeraWulf Inc.	Data Center	Corporate	BB
Wulf Compute, LLC	Data Center	Corporate	BB
Zenith Arc LLC	Data Center	Project Finance	BB
Applied ComputeCo 3 LLC	Data Center	Corporate	BB-
EDC Venture LLC	Data Center	Corporate	BB-
Elk Grove Village Property LLC	Data Center	Project Finance	BB-
Galaxy Helios Data Centers II LLC	Data Center	Corporate	BB-
Meridian Arc Holdco LLC	Data Center	Project Finance	BB-
Applied Digital Corporation	Data Center	Corporate	BB-

As of Oct. 6, 2026.

Unconventional Financing Structures Are Not Uncommon

Main types of circular financing in the AI sector	S&P's Treatment
Residual-value guarantee	Apply debt-like adjustment if guaranteed value exceeds S&P stressed valuation of underlying assets
Chip financing	Calculate chipmaker's exposure by comparing amortized loan value with carrying value of AI chips
Lease guarantee for third-party leases	Apply debt adjustment if ultimate beneficiary of guarantee is not investment-grade
Power purchase agreements	Apply debt adjustment for fixed and minimum component of such agreements
Lease liabilities	Include in debt metrics
Backstop guarantees	Include in debt metrics
Equity investments	Treat as risk amplifier

How A Collapse Of One AI Entity Could Expose Circular Financing Risks

Known knowns

- Steep capex that is outpacing returns on investment.
- Steep depreciation of data center equipment.
- Mismatch between data center leases with customers (3-5 years) and the useful life of the data center (20+ years)

Known unknowns

- Potential for cheap, open-source models to undercut the economics of frontier AI labs.
- When returns on investment on AI will sustainably exceed ongoing capex.
- Degree to which circular financing pushes up the revenues of chipmakers and hyperscalers.

Unknown knowns

- Possibility that a prominent failure in the sector could sink investors' current large appetite to fund everything AI.
- Emergence of losers in the AI race could create ripple effects, magnified by circular financing, undermining confidence needed to sustain the sector's hefty capex.

Unknown unknowns

- Risks we are not yet aware of and can't specifically identify.

Related Research

- Data Center Force Majeure: A Little Room To Breathe Or One Step Closer To The Edge?, Oct. 13, 2026
- Data Center Cyber Risk Is Increasing And Underrecognized, Sept. 24, 2026
- Credit Outlook For Hyperscalers: A Temperature Check, Sept. 3, 2026
- Australian Data Centers: Navigating Growth And Scrutiny, July 26, 2026
- Malaysia's Data Center Outlook: A Reset For Sustainable Growth, July 19, 2026
- Decoding Data Center Risk: A Multi-Dimensional Analysis, July 1, 2026
- Potential RMB2 Trillion Data Center Plan Could Drive China Tech Hardware Demand, June 15, 2026
- Japan Data Centers: Supply Squeeze Will Shape Competition, June 10, 2026
- The Insurance Gap Is Reshaping Hyperscale Data Center Finance, June 08, 2026
- Credit FAQ: Decoding Data Center ABS Utility Scores, June 2, 2026
- North America Data Center ABS Roundup: Second-Quarter 2026, May 28, 2026
- Scenario Analysis: Global Data Center ABS Ratings Are Holding Steady, May 20, 2026
- Project Finance: Digital Infrastructure: A Rapidly Evolving Digital Infrastructure Market Is Blurring Credit Lines, Feb. 23, 2026

Related Research

- ABS Frontiers: Equipping Data Centers Through Securitization, Feb. 19, 2026
- Global Infrastructure And Project Finance: Key Data Center Financing Takeaways From PPIF 2026, Feb 6, 2026
- From Growth To Growing Risk: Rapid Development Of Data Centers Is Creating Vulnerabilities, Jan. 21, 2026
- Credit FAQ: How Does S&P Global Ratings Apply Project Finance Methodology To Analyze Data Centers?, Oct. 6, 2025
- The Four Main Approaches For Rating Data Center Financings, June 13, 2024

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